

· OPENCLAW COURSE · EXTRA CLASS

· Is your agent crashing? This is probably why  
Rate

Limits:

Why it crashes

and how to solve it

Understand the 429, learn about Anthropic and OpenAI tiers,  
and learn practical strategies to never again have  
your agent stopped due to lack of quota.

DURATION

12 min

LEVEL

Intermediate

PRE-REQ

Active API Key

· ERROR LOG

429 Too Many Requests

RateLimitError: Rate limit  
reached for claude-sonnet-  
4-5. Retrying in 32s·

## EXTRA CLASS · RATE LIMITS

[openclaw.ai/corso](https://openclaw.ai/corso)

### 01 What is Rate Limit?

The provider limits how many requests you can make per unit of time.

Rate limit is a protection mechanism that every API provider uses. When you exceed the quota, the API rejects the call with HTTP status 429 · "Too Many Requests". It's not your fault; it is an infrastructure rule that exists for good reasons:

#### REASON 1

- Protect infrastructure

Avoid server overload

Ensure platform stability

Prevent abuse and accidental DDoS

#### REASON 2

- Quality for all

No user monopolizes resources

Fair SLA for the entire base

Tiers reflect usage and commitment

- The 3 types of limit

#### RPM

Requests Per Minute

Number of calls per minute · the most common

#### TPM

Tokens Per Minute

Volume of tokens (input + output) per minute

#### RPD

Requests Per Day

Daily call limit (Free tier)

### 02 How it appears in OpenClaw

The most obvious sign is 429 in the log · but it's not always that clear.

- When it happens more

- Many crons at the same time

ERROR RateLimitError: 429 Too Many Requests

at Anthropic.makeRequest (sdk/client.js:142)

WARN Rate limit reached for model claude-sonnet-4-5

x-ratelimit-remaining-requests: 0

x-ratelimit-remaining-tokens: 0

x-ratelimit-reset-requests: 60s

INFO Retrying in 32s (attempt 1/3)·

LOG

10 scheduled crons for 9:00 AM trigger at the same time · RPM burst in seconds. The agent "freezes" during rush hour.

- Parallel sub-agents

Each sub-agent makes its own calls. Three sub-agents running together = 3× the pressure on the rate limit of the same API Key.

- Very frequent heartbeats

Heartbeat configured every 5 min with heavy model + large context · accumulates TPM quickly throughout the day.

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### 03 Limits per Provider

Know the tiers and know where you are now.

· Anthropic (Claude)

TIER

RPM

TPM

RPD

REQUIREMENT

Free

5

25k

100

Account created

Build Tier 1

50

50k

1k

\$5 spent

Build Tier 2

1,000

160k

·

\$50 spent

Scale Tier 3

2,000

800k

·

\$100 spent

Scale Tier 4

4,000

2.4M

·

\$250 spent

· OpenAI (GPT)

TIER

RPM

TPM

REQUIREMENT

Free

3

40k

Account created

Tier 1

500

200k

\$5 spent

Tier 2

5,000

2M

\$50 spent

Tier 3

10,000

4M

\$100 spent

· How to level up your tier

Tier 4

· \$250

Scale

· Pay attention to the Free tier

5 RPM means that if you have 2 crons running together with 3 calls each, you will already hit the limit in seconds. With a new account, this is the most common scenario of "stuck agent".

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### 04 Strategies in OpenClaw

Four complementary approaches · use the right combination for your usage level.

#### · A) Reduce Pressure

Before trying anything else: spend less quota. It seems obvious, but most problems are solved here.

#### · Minimum model for each task

Haiku for heartbeats and routine crons · uses 10× fewer tokens than Sonnet.

Sonnet for daily interaction and tasks with medium context.

Opus only when you need deep reasoning · never as default.

#### · Space out the crons

Never schedule multiple crons for the exact same time. Distribute with at least 5 minutes interval:

9:00 · daily summary | 9:05 · check emails | 9:10 · calendar sync

#### · Avoid sub-agents in parallel

If a task can be done in sequence, don't trigger it in parallel. Each sub-agent consumes quota from the same API Key. Prefer

#### · B) Retry with Automatic Backoff

OpenClaw already takes care of this for you. When it receives a 429, the runtime waits and retries automatically.

1

API returns 429

Limit reached · call rejected by the provider

2

OpenClaw starts backoff

Exponential waiting: 1s · 2s · 4s · 8s · until the window resets

3

Automatic retry · without intervention

The call is retried. You don't need to do anything.

·A "slow" agent isn't a "stuck" agent

If you notice that the agent took 30-60 seconds longer to respond, it was probably in backoff. This is correct behavior·it's not a failure. The agent will respond; it just needs to wait.

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### · C) Distribute Load

When reducing and waiting aren't enough, the next frontier is distributing work in time and space.

### · Heavy jobs · sessionTarget: isolated

Use sessionTarget: "isolated" for sub-agents that make many calls. This ensures that the heavy job doesn't compete with the main session's interactions for the same rate limit window.

### · D) Real-Time Monitoring

#### RESPONSE HEADER

##### MEANING

##### ACTION IF LOW

x-ratelimit-remaining-requests

Calls remaining in the current window

Reduce

frequency

x-ratelimit-remaining-tokens

Tokens remaining in the current window

Use smaller

model

x-ratelimit-reset-requests

Time until RPM resets

Wait for backoff

x-ratelimit-reset-tokens

Time until TPM resets

Wait for backoff

# · Avoid: 3 crons at the same minute  
crons:

- schedule: "0 9 \* \* \*"  
name: daily-summary
- schedule: "0 9 \* \* \*"  
name: check-emails
- schedule: "0 9 \* \* \*"  
name: sync-calendar

# · Prefer: distributed in 5 min

crons:

- schedule: "0 9 \* \* \*"  
name: daily-summary
- schedule: "5 9 \* \* \*"  
name: check-emails # 09:05
- schedule: "10 9 \* \* \*"  
name: sync-calendar # 09:10

YAML



· Check your tier on the dashboard

Anthropic:

[console.anthropic.com](https://console.anthropic.com) · Settings · Limits |

OpenAI:

[platform.openai.com](https://platform.openai.com) · Settings · Limits

05 Quick Diagnosis

"My agent crashed · is it rate limit?"

· Diagnosis Checklist

Do you have multiple crons scheduled for the same time?

Do you have sub-agents running in parallel in this session?

Does the log mention 429 or RateLimitError ?

Did the agent start working again on its own after a few minutes?

Are you on the

Free or

Tier 1 tier with intense use?

· If you checked 3 or more · almost certainly rate limit

The fastest solution: space out the crons, use Haiku in the heartbeat, and wait for the automatic retry. If it persists, check the

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### 06 Rate Limit and Cost

Clarifies a frequent confusion · they are two different things.

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#### Rate Limit

Limits the speed of calls. When reached, the agent waits. It doesn't increase the cost · waiting tokens don't exist. It only delays execution.

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#### Billing Limit

Limits monthly spending. Configurable on the dashboard. When reached, calls are blocked until the cycle resets or the limit is increased.

· Rate limit doesn't charge you anything extra

You only pay for the tokens that the model processed. The time in backoff does not generate charges. An agent that was in retry for 2 minutes used exactly the same tokens as one that responded in 10 seconds.

### 07 Advanced Tip: Multiple API Keys

For intensive use, distributing between keys (or providers) is the way to go.

· Multiple keys in the same provider

Each API Key has its own independent rate limit. Creating two keys in Anthropic and rotating between them doubles your effective RPM. Works well for predictable workloads.

· Different providers per agent

Anthropic for critical production agents.

OpenAI as backup or for specific code tasks.

OpenClaw natively supports multiple providers · configure distinct tokens per agent.

429

RATE LIMIT · WAIT

402

BILLING LIMIT · PAY

\$0

COST OF TIME

WAITING

# OpenClaw · configure providers per agent

CONFIG

· Practical rule

For 90% of the course users, the rate limit disappears after spending \$5 (Tier 1) and spacing out the crons. Multiple keys and multi-provider are for those who are scaling seriously

· don't jump to that first.

· Anti-Rate-Limit Checklist

Heartbeat using claude-haiku-4-5 (not Sonnet or Opus)

Distributed crons with · 5 min interval between them

Heavy jobs with sessionTarget: "isolated"

Tier verified at [console.anthropic.com](https://console.anthropic.com) · Settings · Limits

Billing limit configured (do not leave unlimited)

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## RELATED CLASS

N-7 · How Much Does It Cost and Which Model to Use

# Main agent uses Anthropic

openclaw config set model anthropic/claude-sonnet-4-5

# Backup agent configured with OpenAI

# (via OPENAI\_API\_KEY in env or config)

openclaw config set model openai/gpt-4o --agent backup-agent

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